6. Project Testing Phase

Functional testing and validation checklist

Testing Objective

Verify that every documented feature accepts the expected input, reaches the correct backend module and returns a usable result.

Test Cases

- TC01 — Q&A;: enter a factual question; expected result: an AI-generated answer is displayed.
- TC02 — Explanation: enter a topic; expected result: a simplified explanation is displayed.
- TC03 — Quiz: enter a topic/passage; expected result: three MCQs with four options each and correct answers are generated.
- TC04 — Summary: enter a long paragraph; expected result: a concise summary is returned.
- TC05 — Learning Path: enter a topic; expected result: structured beginner-to-advanced guidance is returned.
- TC06 — Frontend integration: submit each task through the UI; expected result: response appears below the input.
- TC07 — Error handling: provide an invalid/incomplete AI response scenario; expected result: the application reports an error rather than silently failing.

Evidence to Capture

- Screenshot of the home page.
- Screenshot of a successful Q&A; response.
- Screenshot of a concept explanation.
- Screenshot of generated quiz.
- Screenshot of summary output.
- Screenshot of learning recommendations.
- Screenshot showing the application running locally.

Testing Result Template

For the final submission, record Test Case ID, input, expected output, actual output, status (Pass/Fail), date and screenshot filename.